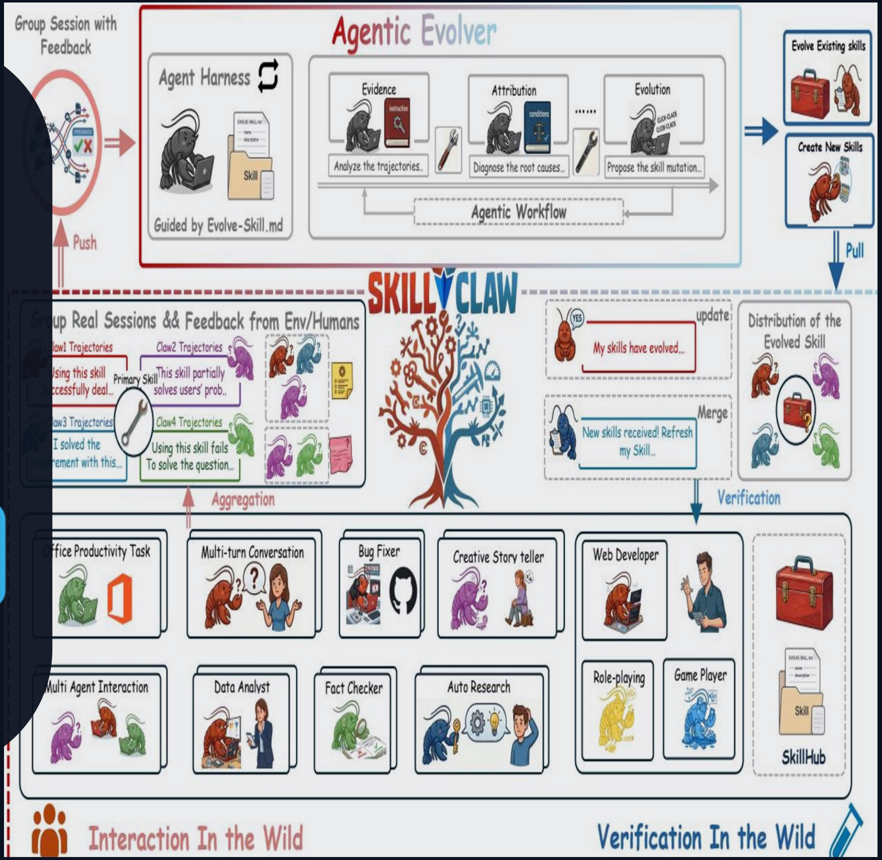
SkillClaw

Collective Skill Evolution for
Multi-User LLM Agent Ecosystems

From cross-user trajectories → verified skill updates



Source: arXiv 2604.08377 (https://arxiv.org/abs/2604.08377)

Agent Skills Remain Static After Deployment

Problem

Observed failure mode in deployed agent ecosystems:

Static skill library

- Skills are manually installed from a hub
- Runtime discoveries rarely persist
- Failures & recoveries repeat across users
- System-level knowledge does not accumulate

What is missing?

A closed loop that:
- captures full action–feedback trajectories
- aggregates evidence across users
- updates skills automatically with verification

→ turns individual fixes into shared improvements

Static: install once → never updates

Evolving: evidence → verified updates → synchronize

Closed-Loop Pipeline: Multi-User → Shared Skill Evolution

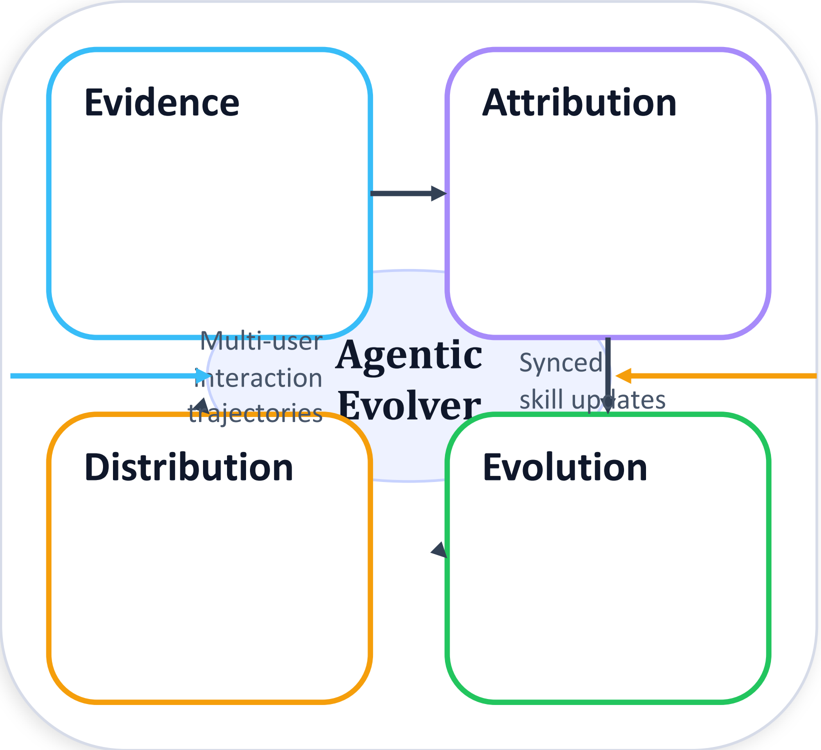
Architecture

Centralized evolution with a closed loop:

- Multi-user interaction produces trajectories
- Sessions grouped by referenced skill
- Agentic evolver proposes mutations
- Verified updates synchronize back to agents

Four-stage closed loop (this slide):

- Evidence
- Attribution
- Evolution
- Distribution



Agentic Evolver: Evidence → Attribution → Evo

Evolver

Inside the evolver (open-ended reasoning; not fixed rules):

1) Evidence

Analyze grouped trajectories
Identify behavioral patterns
Preserve action–feedback chains

2) Attribution

Diagnose root causes
Explain failures vs successes
Pinpoint boundary conditions

3) Evolution

Propose skill mutations
Refine existing or create new
Submit for verification

Cross-User Trajectory Aggregation Reveals Behavioral Boundaries

Aggregation

Why cross-user evidence matters:

Complementary signals

Different users exercise the same skill under diverse contexts, producing:
- success patterns
- failure modes

→ a clearer boundary of when it works/breaks

Single-user limitation

A single user rarely generates enough signal to distinguish:
- generalizable improvement

vs

Compatible Claw-style systems

- OpenClaw
- CoPaw
- IronClaw
- PicoClaw
- ZeroClaw
- NanoClaw
- NemoClaw

Aggregation step:

Group sessions by referenced skill
→ shared evidence base per skill

Case Study: Multimodal Creative Pipeline — 6-Day Evolution

Case Study

Conservative verification: 1 accepted out of 6 candidates

Day	Outcome	Rationale (from evolution log)
1	Accepted	validate-tmp-workspace-inputs: check /tmp_workspace inputs & environment setup
2	Rejected	multimodal-input-validation-and-setup: multimodal input validation & output env init
3	Rejected	multimodal-creative-task-pipeline: multimodal creative pipeline (extract from PDF/video/image)
4	Rejected	Added image classification, visual generation, structured validation
5	Rejected	Creative pipeline + per-file fail-fast validation
6	Rejected	Extended PDF [illegible]

Interpretation: The validator admits updates only when evidence supports a safe improvement, preventing regressions.

Case Study: Git Push Auth-Fallback — 6-Day Refinement

Case Study

Iterative refinement: 3 accepted out of 6 days (then plateau)

Day	Outcome	Rationale (from evolution log)
1	Accepted	Safe fallback instead of blocking on push failure
2	Accepted	Unified patch/bundle filenames; reduced inconsistency
3	Accepted	Auth-alternative paths & secrets audit; fixed subshell pitfalls
4	Accepted	Same-pool retest; validator confirmed current pool as best
5	Rejected	Added “push hang treated as auth failure”; no improvement
6	Rejected	[illegible]

Contrast vs creative pipeline

Higher acceptance rate early (3/6)
then validator rejects non-improving changes

What the plateau indicates

Evidence no longer supports further gains
→ conservative guardrail against churn

Key Takeaways & Implications

Conclusion

Three distinguishing properties:

1) Collective evolution

Individual interactions improve a shared skill ecosystem.

2) Fully automatic

Runtime interaction drives updates without manual curation.

3) Agentic evolution paradigm

Open-ended reasoning over evidence, not predefined rules.